

Differentiate this with respect to y and use (4b) to get

$$\frac{\partial u}{\partial y} = x + \frac{dk}{dy} = N = x - e^{-y}, \quad \frac{dk}{dy} = -e^{-y}, \quad k = e^{-y} + c^*.$$

Hence the general solution is

$$u(x, y) = e^x + xy + e^{-y} = c.$$

Step 3. Particular solution. The initial condition $y(0) = 1$ gives $u(0, -1) = 1 + 0 + e = 3.72$. Hence the answer is $e^x + xy + e^{-y} = 1 + e = 3.72$. Figure 15 shows several particular solutions obtained as level curves of $u(x, y) = c$, obtained by a CAS, a convenient way in cases in which it is impossible or difficult to cast a solution into explicit form. Note the curve that (nearly) satisfies the initial condition.

Step 4. Checking. Check by substitution that the answer satisfies the given equation as well as the initial condition. ■

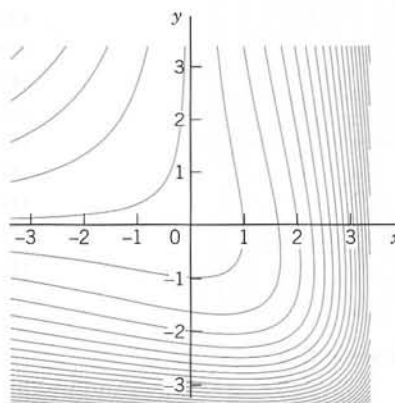


Fig. 15. Particular solutions in Example 5

PROBLEM SET 1.4

1–20 EXACT ODEs. INTEGRATING FACTORS

Test for exactness. If exact, solve. If not, use an integrating factor as given or find it by inspection or from the theorems in the text. Also, if an initial condition is given, determine the corresponding particular solution.

1. $x^3 dx + y^3 dy = 0$
2. $(x - y)(dx - dy) = 0$
3. $-\pi \sin \pi x \sinh y dx + \cos \pi x \cosh y dy = 0$
4. $(e^y - ye^x) dx + (xe^y - e^x) dy = 0$
5. $9x dx + 4y dy = 0$
6. $e^x(\cos y dx - \sin y dy) = 0$
7. $e^{-2\theta} dr - 2re^{-2\theta} d\theta = 0$
8. $(2x + 1/y - y/x^2) dx + (2y + 1/x - x/y^2) dy = 0$
9. $(-y/x^2 + 2 \cos 2x) dx + (1/x - 2 \sin 2y) dy = 0$
10. $-2xy \sin(x^2) dx + \cos(x^2) dy = 0$
11. $-y dx + x dy = 0$
12. $(e^{x+y} - y) dx + (xe^{x+y} + 1) dy = 0$
13. $-3y dx + 2x dy = 0, \quad F(x, y) = y/x^4$
14. $(x^4 + y^2) dx - xy dy = 0, \quad y(2) = 1$
15. $e^{2x}(2 \cos y dx - \sin y dy) = 0, \quad y(0) = 0$
16. $-\sin xy (y dx + x dy) = 0, \quad y(1) = \pi$
17. $(\cos \omega x + \omega \sin \omega x) dx + e^x dy = 0, \quad y(0) = 1$
18. $(\cos xy + x/y) dx + (1 + (x/y) \cos xy) dy = 0$
19. $e^{-y} dx + e^{-x}(-e^{-y} + 1) dy = 0, \quad F = e^{x+y}$
20. $(\sin y \cos y + x \cos^2 y) dx + x dy = 0$
21. Under what conditions for the constants A, B, C, D is $(Ax + By) dx + (Cx + Dy) dy = 0$ exact? Solve the exact equation.

22. CAS PROJECT. Graphing Particular Solutions

Graph particular solutions of the following ODE, proceeding as explained.

$$(21) \quad y \cos x \, dx + \frac{1}{y} \, dy = 0$$

- (a) Test for exactness. If necessary, find an integrating factor. Find the general solution $u(x, y) = c$.
 (b) Solve (21) by separating variables. Is this simpler than (a)?
 (c) Graph contours $u(x, y) = c$ by your CAS. (Cf. Fig. 16.)

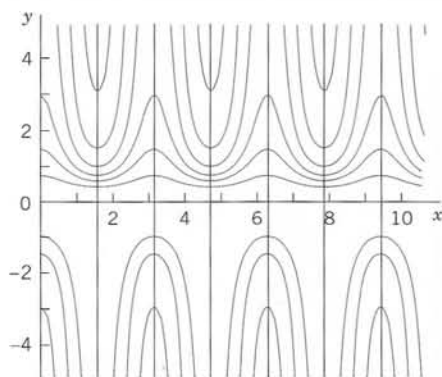


Fig. 16. Particular solutions in CAS Project 22

(d) In another graph show the solution curves satisfying $y(0) = \pm 1, \pm 2, \pm 3, \pm 4$. Compare the quality of (c) and (d) and comment.

(e) Do the same steps for another nonexact ODE of your choice.

23. WRITING PROJECT. Working Backward. Start from solutions $u(x, y) = c$ of your choice, find a corresponding exact ODE, destroy exactness by a multiplication or division. This should give you a feel for the form of ODEs you can reach by the method of integrating factors. (Working backward is useful in other areas, too; Euler and other great masters frequently did it.)

24. TEAM PROJECT. Solution by Several Methods. Show this as indicated. Compare the amount of work.

(A) $e^y(\sinh x \, dx + \cosh x \, dy) = 0$ as an exact ODE and by separation.

(B) $(1 + 2x) \cos y \, dx + dy/\cos y = 0$ by Theorem 2 and by separation.

(C) $(x^2 + y^2) \, dx - 2xy \, dy = 0$ by Theorem 1 or 2 and by separation with $v = y/x$.

(D) $3x^2 y \, dx + 4x^3 \, dy = 0$ by Theorems 1 and 2 and by separation.

(E) Search the text and the problems for further ODEs that can be solved by more than one of the methods discussed so far. Make a list of these ODEs. Find further cases of your own.

1.5 Linear ODEs. Bernoulli Equation. Population Dynamics

Linear ODEs or ODEs that can be transformed to linear form are models of various phenomena, for instance, in physics, biology, population dynamics, and ecology, as we shall see. A first-order ODE is said to be **linear** if it can be written

$$(1) \quad y' + p(x)y = r(x).$$

The defining feature of this equation is that it is linear in both the unknown function y and its derivative $y' = dy/dx$, whereas p and r may be **any** given functions of x . If in an application the independent variable is time, we write t instead of x .

If the first term is $f(x)y'$ (instead of y'), divide the equation by $f(x)$ to get the “**standard form**” (1), with y' as the first term, which is practical.

For instance, $y' \cos x + y \sin x = x$ is a linear ODE, and its standard form is $y' + y \tan x = x \sec x$.

The function $r(x)$ on the right may be a force, and the solution $y(x)$ a displacement in a motion or an electrical current or some other physical quantity. In engineering, $r(x)$ is frequently called the **input**, and $y(x)$ is called the **output** or the **response** to the input (and, if given, to the initial condition).